David Phan

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github.com/davphan

Education

University of Washington, Seattle Sept. 2020 - June 2024
Bachelor of Science in Electrical and Computer Engineering (ECE) GPA: 3.87, Cum Laude

Professional Experience

SDE II - Amazon Sept. 2024 - Present

- Operated Amazon's core highly-available distributed media ingestion service, managing operational tasks such as oncall rotations and peak event preparation.
- Modernized Amazon's content ingestion by adding support for media types such as HEIF, reducing monthly incompatibility ingestion related errors by 32%.
- Enhanced team operations effectiveness by leading monthly bug bash sessions, onboarding new team members to our team oncall rotations, and migrating our team to new task management tooling.
- Migrated legacy bare-metal deployment infrastructure to CDK based AWS cloud infrastructure.

Software Development Engineer Intern - Amazon Jun. 2023 - Sept. 2023

- Spearheaded internal full-stack application project to aid ML research teams in processing datasets.
- Designed and implemented both a React/Typescript front-end and a REST API using AWS API Gateway, Lambda, s3, DynamoDB, and Amazon internal tools.
- Deployed applications through CI/CD frameworks using Git, internal pipeline tools, and AWS Pipelines.
- Collaborated with a team of developers to generate and meet design goals using the Agile Methodology/Framework.

Software Engineering Intern - nFlux.ai Sept. 2021 – Dec. 2021

- Analyzed and processed video files for ML training and worked with senior engineers on DevOps tasks to develop CI/CD pipelines using Docker Compose and Git Actions to enhance project workflow.
- Designed and developed python code that automated and organized database information retrieval for SQL and noSQL databases.
- Developed functionality testing procedures to analyze UI/UX components of company products.

Research Experience

Software Development Intern - Makeability Lab, Project Sidewalk Mar. 2022 – Jun. 2024

- Maintain and update the Project Sidewalk API and website, which tracks and updates geographical information on pedestrian accessibility issues in current city infrastructure.
- Collaborate with a team of software developers to respond to GitHub tickets to quickly debug issues and implement new features in the code base or frontend UI/UX.
- Utilize technologies like Scala, PostgreSQL, and other frontend languages/frameworks.

Undergraduate Researcher - Theberge (BCME) Lab Jul. 2022 – Jun. 2024

- Automated data-analytics and image processing tasks by coding python-based desktop applications.
- Designed and prototyped microfluidic and biomedical devices for research and medical purposes using CAD software, 3D printing and Milling devices, and microcontroller circuits.
- Built and optimized website features to represent lab research and expand study group participation.

Teaching and Mentoring Experience

Undergraduate Electrical and Computer Engineering Tutor Jan. 2024 - Jun. 2024
UW College of Engineering

- Guided students through 11 different introductory and high level ECE coursework (2xx-4xx level courses) employing combinations of teaching strategies such as Dual Coding (vocal and visual teaching).

Engineering Embassador Mar. 2022 - Jun. 2023
UW College of Engineering

- Led future students towards a STEM pathway by guiding them towards different areas of interest, creating engaging presentations and activities.
- Coordinated events by working in small teams and reporting out in larger conference meetings.
- Analyzed demographic data and statistics in engineering to understand and act on diversity issues.

Publications

TK TK

- TK

TK TK

- TK

Other Experience

HackUPC Hackathon May 2021

- Developed front end webpage for a facial recognition security system using HTML, CSS, and the Bootstrap 4 framework.

UW Hyperloop Propulsions Team Oct. 2020 – Jul. 2021

- Developed coil-windings and prototype CAD models for a Dual-Sided Linear Induction Motor (DSLIM), utilizing independent self-guided research and team communication to design UW's first DSLIM.
- Led planning and communications for manufacturing outsourcing of material components.

Washington Aerospace Scholars (WAS) Nov. 2017 – Aug. 2019

- Designed a simulated manned Mars mission using skills from mission design to fundamental engineering practices learned through a rigorous multi-year long aerospace engineering curriculum.
- Led project management between sub-teams as the propulsions team's Communications Expert and won the William E. Boeing award for leadership.

Pharmacy Technician-in-Training – Walgreens Jun. 2021 – Sept. 2021

- Performed pharmacy technician duties and fostered customer relations behind the counter by problem solving issues and showing exceptional customer service.

Technical Skills

Programming languages/software: Java, Python, Bash/Command Line scripting, C, C++, Jupyter Notebook, Scala, VS Code, HTML, CSS, Javascript/Typescript, React, Bootstrap 4, Dagger DI, Mockito
Version control/Workflow optimization software: Git, Github, Git Actions, Docker Compose
Database and Cloud service software: AWS, Azure, MongoDB, PostgreSQL
Leadership/Project Management: Engineering Ambassador at UW, ECE Tutor at UW, Officer for Music for Charity at UW, School Music Director, Communications Lead at Washington Aerospace Scholars (WAS)

References

Shilpa Gupta, SDE Manager
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Ashleigh Theberge, Professor
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Jon Froelich, Professor
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Mikey Stausged, TK
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